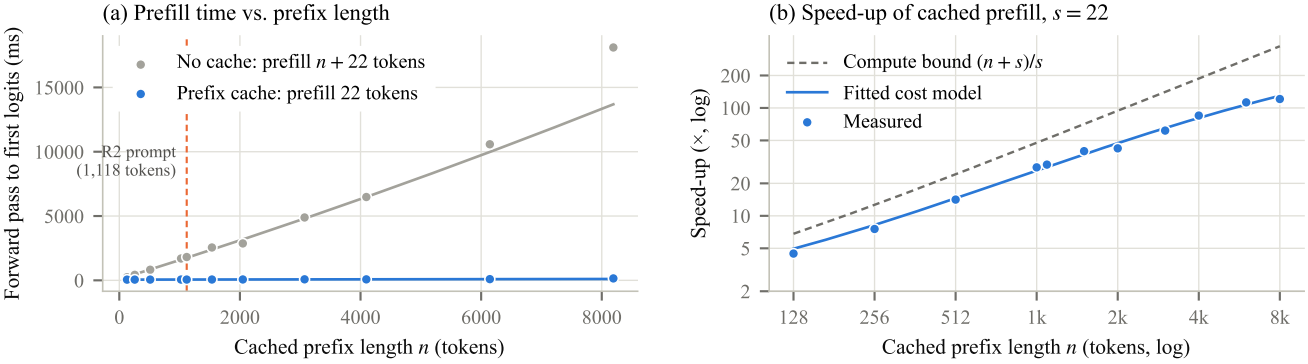
(a) Prefill time vs. prefix length
No cache: prefill $n + 22$ tokens
Prefix cache: prefill 22 tokens
R2 prompt
(1,118 tokens)
Forward pass to first logits (ms)
0
5000
10000
15000
0
2000
4000
6000
8000
Cached prefix length n (tokens)
(b) Speed-up of cached prefill, $s = 22$
Compute bound $(n + s)/s$
Fitted cost model
Measured
Speed-up (×, log)
2
5
10
20
50
100
200
128
256
512
1k
2k
4k
8k
Cached prefix length n (tokens, log)